Cloned States, Linear Regret: A Representation-Invariant Theory of Kernel MDPs

Pahan Dewasurendra
Johns Hopkins University

Abstract

A common kernel-MDP assumption requires every scalar transition slice $z \mapsto P_h(s' \mid z)$ to have bounded norm in a state-action RKHS. We show that this condition measures the names assigned to next states, not the decision problem. Replacing a state by m behaviorally identical clones divides every associated slice norm by m, while preserving all policy values, every learning algorithm's attainable observation laws, minimax regret, and the kernel's maximum information gain. More strongly, we construct a fixed two-step MDP domain and a fixed Matérn kernel with sublinear information gain for which the coordinatewise condition, with any prescribed constant, permits linear minimax regret. This gives a negative answer to the COLT 2024 no-regret open problem under its stated assumption. We then introduce the *Bellman radius*, the operator norm of the transition kernel from bounded values to the state-action RKHS. It is invariant to state cloning, is the exact constant needed for Bellman closure, and is equivalent up to a factor two to the semivariation of an RKHS-valued transition measure. An integrable RKHS norm of a transition density is a verifiable stronger condition. It exposes the missing state-volume factor in the frequently cited coordinatewise argument. These results separate valid Bellman-closure and conditional-embedding theorems from representation-dependent sufficient conditions, and give a drop-in premise for future kernel RL guarantees.

1 Introduction

Kernel methods offer a natural route from linear reinforcement learning to nonlinear function approximation. A widely used model assumes that for every next state s', the scalar function $z \mapsto P_h(s' \mid z)$ belongs to the unit ball of a known reproducing kernel Hilbert space (RKHS) on state-action pairs (Yang et al., 2020; Yeh et al., 2023; Vakili & Olkhovskaya, 2023; Vakili et al., 2024; Vakili & Olkhovskaya, 2024; Kayal et al., 2025). The intended consequence is that every Bellman image

$$[P_h V](z) = \int_{\mathcal{S}} V(s') P_h(ds' \mid z) \tag{1}$$

also has controlled RKHS norm. Kernel ridge confidence sets can then be applied to the moving value targets produced by value iteration.

There is a basic obstruction. Split one next-state label into m labels with the same reward and future dynamics, and allocate one mth of its probability to each. The agent has not received a new control problem. It only observes an independent uniform label after the transition. Yet all m scalar probability functions now have one mth of the original norm. This operation can make the coordinate bound arbitrarily small without making learning easier.

The issue is not merely aesthetic. We use state cloning to encode an unrestricted smooth continuum-armed bandit inside transition probabilities whose individual RKHS norms are arbitrarily small. The construction uses one fixed countable state space and one fixed Matérn

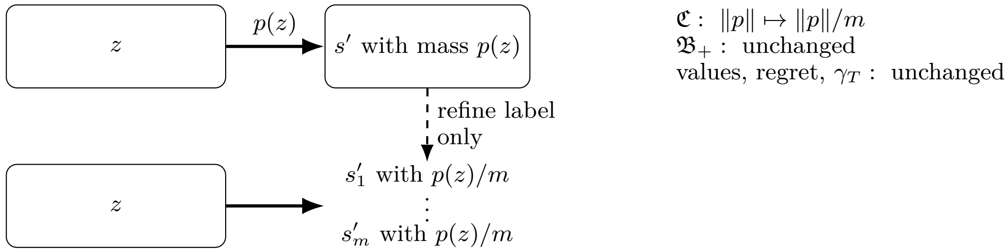

Figure 1: State cloning changes a coordinatewise RKHS parameter without changing the learning problem. Clone indices are independent random labels.

kernel. Its maximum information gain is sublinear, but its minimax regret is linear. Thus the coordinatewise premise in the kernel-RL open problem of Vakili (2024) cannot by itself support no-regret learning.

The right object is already implicit in Bellman's equation. We define

$$\mathfrak{B}_+(P; \mathcal{H}_k) = \sup_{V:\, 0 \le V \le 1} \|PV\|_{\mathcal{H}_k}. \tag{2}$$

This *Bellman radius* is unchanged by cloning and is exactly the smallest constant controlling every bounded nonnegative value image. For discrete states, its signed counterpart is the semivariation of the RKHS-valued measure $A \mapsto P(A \mid \cdot)$. For transition densities $p(s' \mid \cdot)$ relative to a reference measure μ, the stronger quantity $\int \|p(s' \mid \cdot)\|_{\mathcal{H}_k}\, d\mu(s')$ controls the radius. Unlike a pointwise density bound, this integral retains the state-volume factor and is invariant to a change of density convention.

Contributions.

- We prove an exact state-refinement theorem. State cloning preserves the entire projected learning experiment, minimax regret, and maximum information gain, while coordinate transition norms decrease by arbitrary factors.
- We prove a linear minimax-regret lower bound under the stated assumption of Vakili (2024). The state space, reward, and Matérn kernel are fixed, and the kernel has $o(T)$ information gain.
- We characterize a representation-invariant replacement through the Bellman radius, semivariation, and transition-density variation. The hierarchy gives exact Bellman closure and clarifies when coordinate assumptions can be repaired.
- We audit the proof-level consequence across kernel-RL results from 2020 to 2026. Direct Bellman-closure, smooth-Bellman, and bounded conditional-embedding results remain intact. Claims that derive closure from coordinate slices require a cardinality, volume, or operator-radius term.

2 Coordinate norms are not properties of an MDP

Let P be a Markov kernel from a current state-action domain $\mathcal{Z}$ to a finite or countable next-state space $\mathcal{S}$. Let k be a kernel on $\mathcal{Z}$ with RKHS $\mathcal{H}_k$. Write $p_y = P(\{y\} \mid \cdot)$ and define the coordinate radius

$$\mathfrak{C}(P; \mathcal{H}_k) = \sup_{y \in \mathcal{S}} \|p_y\|_{\mathcal{H}_k}. \tag{3}$$

This is the premise in Vakili (2024), up to a constant u.

Choose integers $m_y \ge 1$. The refined state space is $\widetilde{\mathcal{S}} = \{(y,i) : y \in \mathcal{S},\ i \in [m_y]\}$ and the refined transition is

$$\widetilde{P}((y,i) \mid z) = \frac{P(y \mid z)}{m_y}. \tag{4}$$

For a full MDP, rewards and future transitions are constant within each fiber. If current states are also refined, use the pulled-back kernel $\widetilde{k}((s,i,a),(s',j,a')) = k((s,a),(s',a'))$.

Theorem 1 (Exact state refinement)**.** *For any episodic MDP and any finite clone counts, the refined MDP has the following properties. The regret statement applies to any model family and its paired family of refinements.*

1. *Projection of a refined trajectory onto original state labels has the original trajectory law. Conversely, every policy using clone labels can be simulated in the original MDP using private randomization. Hence paired policies have identical value and regret laws, and paired model families have the same minimax regret for every number of episodes.*

2. *The pulled-back RKHS is isometric to the original RKHS and*

$$\mathfrak{C}(\widetilde{P}; \widetilde{\mathcal{H}_k}) = \sup_{y \in \mathcal{S}} \frac{\|p_y\|_{\mathcal{H}_k}}{m_y}. \tag{5}$$

3. *For every regularization $\lambda > 0$ and budget T, the maximum information gain satisfies $\gamma_T(\widetilde{k}; \lambda) = \gamma_T(k; \lambda)$.*

The proof couples clone indices as independent uniforms and observes that every refined Gram matrix equals the Gram matrix of its projection. Full details, including history-dependent randomized policies, are in Appendix B. The result is an exact bisimulation-style refinement (Givan et al., 2003), strengthened here to equality of the sequential learning experiments and kernel complexity.

Corollary 2 (Vacuity under finite refinement)**.** *Let a finite MDP have a finite state-action domain and let k be strictly positive definite on that domain. For every $\epsilon > 0$, there is a behaviorally equivalent refinement with a pulled-back kernel satisfying $\mathfrak{C}(\widetilde{P}_h; \widetilde{\mathcal{H}_k}) \le \epsilon$ for every stage h. Rewards keep their RKHS norms and every γ_T is unchanged.*

Every function on a finite domain belongs to the RKHS of a strictly positive definite Gram matrix. Choosing m_y large enough proves the corollary. Therefore a uniform coordinate bound can certify every finite decision problem after harmless state refinement. The next result rules out the objection that the pulled-back kernel is responsible.

3 A fixed Matérn kernel still permits linear regret

We consider the online episodic setting of Vakili (2024). Rewards are deterministic and known, transitions are unknown, and regret after T episodes is $R_T = \sum_{t=1}^{T}(V_1^* - V_1^{\pi_t})$. The assumption is that a known kernel k satisfies $\|P_h(s' \mid \cdot)\|_{\mathcal{H}_k} \le u$ for every h and s', where $u > 0$ is fixed.

Theorem 3 (No uniform no-regret under coordinate smoothness)**.** *Fix any $u > 0$ and any Matérn smoothness $\nu > 1/2$. There exist a fixed compact state-action domain $\mathcal{Z}$, a fixed countable state space, a fixed known reward, and the restriction to $\mathcal{Z}$ of a unit-variance Matérn kernel on $[0,1]^2$ such that*

$$\inf_{\mathsf{A}} \sup_{M :\, \mathfrak{C}(P_h; \mathcal{H}_k) \le u} \mathbb{E}_M[R_T(\mathsf{A})] \ \ge c_\nu T \qquad \textit{for every } T \ge 2, \tag{6}$$

where $c_\nu > 0$ does not depend on T. At the same time, $\gamma_T(k;1) = \widetilde{O}(T^{2/(2\nu+2)}) = o(T)$. The claim already holds for horizon two.

Construction and proof idea. At the first stage there is one state and the action is $x \in [0,1]$. Terminal states are fixed labels $\{g_n, b_n : n \in \mathbb{N}\}$, with known rewards α and zero. For $M = 2T$, place smooth unit-height bumps $\psi_1, \ldots, \psi_M$ on disjoint intervals. Instance j has total probability

$$f_j(x) = \tfrac{1}{2} + \Delta \psi_j(x) \tag{7}$$

of reaching a good state. Spread this mass uniformly across N_T good labels and the complementary mass across N_T bad labels. Each scalar transition function is therefore f_j/N_T

or $(1-f_j)/N_T$. Smooth compactly supported functions belong to every finite-smoothness Matérn RKHS on a compact domain (Wendland, 2005; Kanagawa et al., 2018). A finite N_T can consequently make every slice norm at most u, even though the Bellman image remains αf_j.

Under the flat instance $f_0 = 1/2$, a length-T action sequence can enter at most T of the $2T$ disjoint supports. A uniformly random support is never entered with probability at least 1/2. Until entry, the complete labeled observation history has the same distribution under f_j and f_0. Hence some j is never entered with probability at least 1/2 under its own instance. On that event all T actions have gap $\alpha\Delta$, proving $\mathbb{E}R_T \geq \alpha\Delta T/2$. Appendix C gives the formal coupling, one common Matérn-domain embedding for all stages, and the randomized-algorithm argument. The standard Matérn information-gain bound gives the final claim (Vakili et al., 2021; Janz et al., 2020).

The hard MDP may depend on T, as is standard in minimax regret. The state space, kernel, reward, horizon, and coordinate constant do not. The number of labels with positive transition mass grows with T. This is precisely the degree of freedom left uncontrolled by the coordinate assumption. Theorem 3 gives a negative answer to the no-regret part of the COLT 2024 open problem *under its stated Assumption 1.* It does not rule out no-regret guarantees under direct Bellman closure, a fixed finite cardinality, or smooth transition densities with controlled integrated norm.

4 The representation-invariant quantity

For a transition kernel P whose relevant Bellman images belong to $\mathcal{H}_k$, define the positive and signed Bellman radii

$$\mathfrak{B}_+(P;\mathcal{H}_k) = \sup_{0\leq V\leq 1} \|PV\|_{\mathcal{H}_k}, \qquad \mathfrak{B}_\pm(P;\mathcal{H}_k) = \sup_{\|V\|_\infty\leq 1} \|PV\|_{\mathcal{H}_k}. \tag{8}$$

For countable states, also define the RKHS slice variation

$$\mathfrak{V}(P;\mathcal{H}_k) = \sum_{y\in\mathcal{S}} \|p_y\|_{\mathcal{H}_k}. \tag{9}$$

Theorem 4 (Radius hierarchy and invariance). *Whenever the quantities are finite,*

$$\mathfrak{C}(P;\mathcal{H}_k) \leq \mathfrak{B}_+(P;\mathcal{H}_k) \leq \mathfrak{B}_\pm(P;\mathcal{H}_k) \leq 2\mathfrak{B}_+(P;\mathcal{H}_k) \leq 2\mathfrak{V}(P;\mathcal{H}_k). \tag{10}$$

Both Bellman radii and the slice variation are exactly invariant under arbitrary state-dependent cloning. The positive radius is the smallest B such that $\|PV\|_{\mathcal{H}_k} \leq B\|V\|_\infty$ for every nonnegative bounded V. The signed radius is the semivariation of the $\mathcal{H}_k$-valued transition measure $A \mapsto P(A \mid \cdot)$ whenever this set function is countably additive in $\mathcal{H}_k$ (Diestel & Uhl, 1977). No reverse inequality in equation (10) *is dimension free.*

The invariance proof is exact. For a function $\widetilde{V}$ on clones, let $\bar{V}(y) = m_y^{-1}\sum_i \widetilde{V}(y,i)$. Then $\widetilde{P}\widetilde{V} = P\bar{V}$, and every V has a constant-on-fibers lift. The variation identity follows by summing m_y copies of $\|p_y/m_y\|_{\mathcal{H}_k}$. The factor-two equivalence uses $V = V_+ - V_-$ for signed V. Examples in Appendix D show strict separations of order n and $\sqrt{n}$ between the three quantities.

The hierarchy also clarifies transition densities. Suppose $P(ds' \mid z) = p(s' \mid z)\mu(ds')$ and $s' \mapsto p(s' \mid \cdot)$ is Bochner integrable in $\mathcal{H}_k$.

Proposition 5 (Density variation). *Define*

$$\mathfrak{V}_\mu(P;\mathcal{H}_k) = \int_{\mathcal{S}} \|p(s' \mid \cdot)\|_{\mathcal{H}_k}\, d\mu(s'). \tag{11}$$

Then $\mathfrak{B}_+(P;\mathcal{H}_k) \leq \mathfrak{V}_\mu(P;\mathcal{H}_k)$. If $\|p(s' \mid \cdot)\|_{\mathcal{H}_k} \leq u$ pointwise, the valid conclusion is $\mathfrak{B}_+(P;\mathcal{H}_k) \leq u\mu(\mathcal{S})$. The quantity in equation (11) *is unchanged under equivalent changes of reference measure and measurable relabelings.*

The proof is the Bochner triangle inequality. Under $d\mu' = w, d\mu$, the new density is $p' = p/w$, so the integrated norm is unchanged. A pointwise density constant alone is therefore incomplete without the reference-measure mass. This matches the general $L^\infty \to \mathcal{H}_k$ conditional-expectation criterion noted by Hertel et al. (2026). Their broader operator theory treats L^p domains, density regularity, and conditional mean embeddings. Our Bellman radius is the exact intrinsic norm for bounded RL value functions.

Corollary 6 (Exact Bellman closure)**.** *If $\|r_h\|_{\mathcal{H}_k} \le B_r$ and $\mathfrak{B}_+(P_h; \mathcal{H}_k) \le B_P$, then for every $V : \mathcal{S} \to [0, H-h]$,*

$$\|r_h + P_h V\|_{\mathcal{H}_k} \le B_r + (H-h)B_P. \tag{12}$$

Conversely, with $r_h = 0$, the smallest uniform constant in equation (12) *is $(H-h)\mathfrak{B}_+(P_h; \mathcal{H}_k)$.*

This is the nonlinear analogue of the integrated vector-measure bound imposed in linear MDPs (Jin et al., 2020). It is also the premise actually needed when a kernel ridge confidence theorem is applied to Bellman targets. Any proof using coordinate smoothness only through this target-norm step can replace it by Corollary 6, with the confidence radius scaled by $B_r + HB_P$.

5 What changes in existing kernel RL theory

Table 1 separates three assumptions that are often described with similar language. They are not equivalent.

Work	Formal premise or proof step	Consequence of our results
Yang et al. (2020)	Main theorem assumes bounded Bellman images directly. A separate paragraph says coordinate slices suffice.	The theorem under direct closure is unchanged. The stated sufficient condition needs slice variation or a finite volume factor.
Yeh et al. (2023)	Coordinate density norm at most one. Lemma 3 proves $\|PV\| \le c\|V\|_\infty$, where $c = \int_{\mathcal{S}} ds'$.	The proof correctly retains state volume c. Omitting c in later uses is not representation invariant.
Vakili et al. (2024)	Lemma 2.3 invokes the same calculation for values on the Lebesgue unit cube.	This normalized specialization is valid because the reference volume is one.
Vakili & Olkhovskaya (2023; 2024); Vakili (2024)	Coordinate masses or densities are bounded, followed by a dimension-free Bellman-target claim.	That implication fails on general state spaces. Theorem 3 rules out uniform no-regret under the open problem's stated premise.
Kayal et al. (2025); Cioba et al. (2026); Pavlovic et al. (2026)	Recent uses cite the same coordinate-to-Bellman implication with an $O(H)$ radius.	The hidden constant must include volume, variation, or $\mathfrak{B}_+$. Other parts of the analyses may remain applicable after this replacement.
Maran et al. (2024a;b)	Smoothness or local linearity is imposed on Bellman images. Appendix F of the latter identifies a separate localization issue in Vakili & Olkhovskaya (2023).	These structural premises are clone invariant and are not targeted by our lower bound.
Chowdhury & Oliveira (2023); Kumar (2025)	A bounded conditional mean embedding operator is assumed. The latter work enforces a state-RKHS ball by projection.	These are operator-level, representation-invariant alternatives, though they control a specified state-function RKHS rather than all bounded values.

Table 1: Proof-level audit. “Unchanged” refers only to the issue studied here. All original auxiliary conditions are still required.

The most precise source for the frequently cited calculation is Yeh et al. (2023, Lemma 3). Its proof integrates the pointwise RKHS norm and obtains $c = \int_{\mathcal{S}} ds'$, exactly as Proposition 5 predicts. For a counting measure, $c = |\mathcal{S}|$. Under m-fold cloning the pointwise coordinate norm falls by m and c grows by m, leaving the product unchanged. Treating c as a universal constant removes the compensating term.

There are two clean routes forward. One can assume the Bellman radius directly, which is minimal for bounded value iteration. Alternatively, one can verify a stronger integrated density or conditional-operator condition. Recent work on conditional expectations provides analytic density criteria (Park & Muandet, 2020; Hertel et al., 2026), while CME-based RL uses a bounded operator between chosen RKHSs (Chowdhury & Oliveira, 2023; Kumar, 2025). The recent projection approach enforces membership in its chosen state-RKHS ball. It does not analyze coordinate slices or all bounded value functions. Smooth-MDP analyses impose regularity on the Bellman operator itself (Maran et al., 2024a;b). Kernel smoothing of a transition distribution is another distinct model and does not rely on coordinate masses (Domingues et al., 2021).

6 Discussion and limitations

State representations are rarely canonical. Equivalent states can arise from tokenization, redundant simulator variables, duplicated observations, or a refined discretization. A structural complexity parameter should not improve merely because such labels are duplicated. Coordinatewise probability masses fail this test. Pointwise transition *densities* can still be meaningful when a canonical reference measure and its volume are part of the model. The distinction must be explicit.

Our lower bound is uniform minimax rather than a single fixed instance with linear asymptotic regret. It exploits an unbounded number of active clone labels, which is allowed by the stated premise. Fixed finite state cardinality restores a factor $|\mathcal{S}|$ and does not contradict tabular learning. We do not propose a new RL algorithm, and a small Bellman radius may itself be difficult to verify from data. The result instead identifies the weakest norm premise needed by a broad family of kernel value-iteration proofs and supplies verifiable stronger conditions.

The conclusion is narrow but consequential. Sublinear kernel information gain describes how well fixed RKHS functions can be learned. It cannot guarantee that the moving Bellman targets have bounded RKHS norm. That second property belongs to the transition operator, not to its individual coordinates.

Ethics statement

This theoretical work uses no human-subject data and introduces no direct societal risk. Its audit of prior claims is phrased at the level of assumptions and proof steps. We distinguish unaffected theorems from affected sufficient conditions.

Reproducibility statement

Appendices B–F contain complete proofs and the fixed-domain lower-bound construction. The supplementary artifact verifies cloning identities, the radius hierarchy on finite kernels, and the hard-instance coupling by exact enumeration. It uses only standard Python and NumPy.

References

Sayak Ray Chowdhury and Rafael Oliveira. Value function approximations via kernel embeddings for no-regret reinforcement learning. In *Proceedings of the 14th Asian Conference on Machine Learning*, volume 189 of *Proceedings of Machine Learning Research*, pp. 249–264. PMLR, 2023.

Alexandru Cioba, Aya Kayal, Laura Toni, Sattar Vakili, and Alberto Bernacchia. Reinforcement learning using known invariances. *arXiv preprint arXiv:2511.03473*, 2026.

Joseph Diestel and Jerry J. Uhl. *Vector Measures*. American Mathematical Society, 1977.

Omar Darwiche Domingues, Pierre Ménard, Matteo Pirotta, Emilie Kaufmann, and Michal Valko. Kernel-based reinforcement learning: A finite-time analysis. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pp. 2783–2792. PMLR, 2021.

Robert Givan, Thomas Dean, and Matthew Greig. Equivalence notions and model minimization in Markov decision processes. *Artificial Intelligence*, 147(1–2):163–223, 2003.

Maximiliano Hertel, Ilja Klebanov, Manuel Schaller, and Karl Worthmann. Verifiable regularity criterion for conditional expectation operators and conditional mean embeddings with applications to nonparametric regression, bayesian inverse problems, and koopman operators. *arXiv preprint arXiv:2608.06155*, 2026.

David Janz, David R. Burt, and Javier González. Bandit optimisation of functions in the Matérn kernel RKHS. In *Proceedings of the 23rd International Conference on Artificial Intelligence and Statistics*, volume 108 of *Proceedings of Machine Learning Research*, pp. 2486–2495. PMLR, 2020.

Chi Jin, Zhuoran Yang, Zhaoran Wang, and Michael I. Jordan. Provably efficient reinforcement learning with linear function approximation. In *Proceedings of the 33rd Conference on Learning Theory*, volume 125 of *Proceedings of Machine Learning Research*, pp. 2137–2143. PMLR, 2020.

Motonobu Kanagawa, Philipp Hennig, Dino Sejdinovic, and Bharath K. Sriperumbudur. Gaussian processes and kernel methods: A review on connections and equivalences. *Foundations and Trends in Machine Learning*, 11(1):1–141, 2018.

Aya Kayal, Sattar Vakili, Laura Toni, and Alberto Bernacchia. Near-optimal sample complexity in reward-free kernel-based reinforcement learning. In *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics*, volume 258 of *Proceedings of Machine Learning Research*, pp. 874–882. PMLR, 2025.

Pawan Kumar. On kernel RL and optimistic closure. OpenReview preprint, forum BoFQUFmXIz, 2025. URL `https://openreview.net/forum?id=BoFQUFmXIz`.

Davide Maran, Alberto Maria Metelli, Matteo Papini, and Marcello Restelli. Local linearity: The key for no-regret reinforcement learning in continuous MDPs. In *Advances in Neural Information Processing Systems*, volume 37, 2024a.

Davide Maran, Alberto Maria Metelli, Matteo Papini, and Marcello Restelli. No-regret reinforcement learning in smooth MDPs. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 34760–34789. PMLR, 2024b.

Junhyung Park and Krikamol Muandet. A measure-theoretic approach to kernel conditional mean embeddings. In *Advances in Neural Information Processing Systems*, volume 33, 2020.

Nikola Pavlovic, Sattar Vakili, and Qing Zhao. Learning kernel-based MDPs from episodic preferential feedback. *arXiv preprint arXiv:2605.23650*, 2026.

Sattar Vakili. Open problem: Order optimal regret bounds for kernel-based reinforcement learning. In *Proceedings of the 37th Conference on Learning Theory*, volume 247 of *Proceedings of Machine Learning Research*, pp. 5340–5344. PMLR, 2024.

Sattar Vakili and Julia Olkhovskaya. Kernelized reinforcement learning with order optimal regret bounds. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Sattar Vakili and Julia Olkhovskaya. Kernel-based function approximation for average reward reinforcement learning: An optimist no-regret algorithm. In *Advances in Neural Information Processing Systems*, volume 37, 2024.

Sattar Vakili, Kia Khezeli, and Victor Picheny. On information gain and regret bounds in gaussian process bandits. In *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics*, volume 130 of *Proceedings of Machine Learning Research*, pp. 82–90. PMLR, 2021.

Sattar Vakili, Farhang Nabiei, Da-Shan Shiu, and Alberto Bernacchia. Reward-free kernel-based reinforcement learning. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 48938–48956. PMLR, 2024.

Holger Wendland. *Scattered Data Approximation*. Cambridge University Press, 2005.

Zhuoran Yang, Chi Jin, Zhaoran Wang, Mengdi Wang, and Michael I. Jordan. Provably efficient reinforcement learning with kernel and neural function approximations. In *Advances in Neural Information Processing Systems*, volume 33, pp. 13903–13916, 2020.

Sing-Yuan Yeh, Fu-Chieh Chang, Chang-Wei Yueh, Pei-Yuan Wu, Alberto Bernacchia, and Sattar Vakili. Sample complexity of kernel-based Q-learning. In *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics*, volume 206 of *Proceedings of Machine Learning Research*. PMLR, 2023.

A Notation and technical preliminaries

For an episodic MDP, $h \in [H]$ indexes the stage, $s \in \mathcal{S}$ the state, $a \in \mathcal{A}$ the action, and $z = (s, a) \in \mathcal{Z}$. Rewards lie in $[0, 1]$. Policies and learning algorithms may be randomized and history dependent. The kernel k is positive semidefinite and has RKHS $\mathcal{H}_k$. Its maximum information gain at regularization $\lambda > 0$ is

$$\gamma_T(k; \lambda) = \sup_{z_1,\dots,z_T \in \mathcal{Z}} \frac{1}{2} \log\det(I_T + \lambda^{-1} K_T), \qquad [K_T]_{ij} = k(z_i, z_j). \tag{13}$$

We allow repeated points in the supremum. This convention has no effect on the arguments.

We use the following elementary pullback fact.

Lemma 7 (Pullback RKHS). *Let $q : \widetilde{\mathcal{Z}} \to \mathcal{Z}$ be surjective and let $\widetilde{k}(\tilde{z}, \tilde{z}') = k(q(\tilde{z}), q(\tilde{z}'))$. Then*

$$\widetilde{\mathcal{H}_k} = \{f \circ q : f \in \mathcal{H}_k\}, \qquad \|f \circ q\|_{\widetilde{\mathcal{H}_k}} = \|f\|_{\mathcal{H}_k}. \tag{14}$$

Also $\gamma_T(\widetilde{k}; \lambda) = \gamma_T(k; \lambda)$.

Proof. If $\phi : \mathcal{Z} \to \mathcal{F}$ is a feature map for k, then $\widetilde{\phi} = \phi \circ q$ is a feature map for $\widetilde{k}$. Surjectivity means the closed linear spans of $\phi(\mathcal{Z})$ and $\widetilde{\phi}(\widetilde{\mathcal{Z}})$ coincide. Their canonical feature-space representations therefore give equation (14). Every T-point Gram matrix in the refined domain is the Gram matrix of its projected points. Conversely, every original sequence has a lift because q is surjective. The two suprema in equation (13) are equal. □

B Proofs for exact state refinement

We first state the full MDP construction. For clarity, take a layered MDP, so a state cannot occur at two different stages. This loses no generality because the stage can be appended to the state. For every $s \in \mathcal{S}_h$, choose a finite integer $m_s \geq 1$ and let

$$\widetilde{\mathcal{S}}_h = \{(s, i) : s \in \mathcal{S}_h,\ i \in [m_s]\}. \tag{15}$$

If the original initial distribution is ρ, set $\widetilde{\rho}(s,i) = \rho(s)/m_s$. Define

$$\widetilde{r}_h((s,i),a) = r_h(s,a), \tag{16}$$

$$\widetilde{P}_h((s',j) \mid (s,i),a) = \frac{P_h(s' \mid s,a)}{m_{s'}}. \tag{17}$$

The projection $q(s,i) = s$ is a probabilistic bisimulation map.

B.1 Equality of decision and learning experiments

Given an original policy π, define its lift by ignoring clone indices. Summing equation (17) over j proves inductively that the projected trajectory has the original law. Rewards agree pathwise under projection.

The converse must account for a refined policy that uses clone indices. An agent in the original MDP can generate them privately. At the initial state s_1, draw I_1 uniformly from $[m_{s_1}]$. After observing each subsequent original state s_h, draw I_h independently and uniformly from $[m_{s_h}]$. Feed the synthetic history $((s_1, I_1), a_1, \ldots, (s_h, I_h))$ to the refined policy. Conditional on the projected history, equation (17) makes the actual clone index in the refined MDP exactly independent and uniform. Induction therefore couples the full reward and projected-state laws.

The same simulation applies across episodes and to a learning algorithm whose next policy depends on all earlier observations. One direction projects observations. The other supplies private clone indices. Thus the sets of attainable reward laws are identical. Optimal values are identical as a special case. Since regret is a function of the selected-policy values and optimal values, the minimax regret is identical for every interaction budget. This proves part 1 of Theorem 1.

B.2 Coordinate norm and information gain

At a transition into s', each refined coordinate function is

$$\widetilde{p}_{s',j}((s,i),a) = \frac{1}{m_{s'}} p_{s'}(s,a). \tag{18}$$

Lemma 7 and homogeneity of the RKHS norm give

$$\|\widetilde{p}_{s',j}\|_{\widetilde{\mathcal{H}_k}} = \frac{\|p_{s'}\|_{\mathcal{H}_k}}{m_{s'}}. \tag{19}$$

Taking the supremum proves equation (5). The information-gain claim is Lemma 7. This completes the proof of Theorem 1.

B.3 Proof of Corollary 2

On a finite domain, strict positive definiteness means the Gram matrix is invertible. Every scalar function f belongs to the RKHS and has finite norm $\|f\|^2_{\mathcal{H}_k} = f^\top K^{-1} f$. There are finitely many stages and target states. Choose

$$m_y \geq \max_h \frac{\|P_h(y \mid \cdot)\|_{\mathcal{H}_k}}{\epsilon} \tag{20}$$

and round upward. Theorem 1 gives the coordinate bound and all invariances. A pulled-back reward has the same norm by Lemma 7.

C Proof of the linear-regret theorem

This section gives one domain and one kernel shared by every interaction budget. The additional geometric detail ensures that all transition coordinates, including their values on state-action pairs that are unreachable at a given stage, are restrictions of Matérn RKHS functions.

C.1 A fixed compact state-action domain

Let

$$\mathcal{S} = \{s_0, g_\infty, b_\infty\} \cup \{g_n, b_n : n \in \mathbb{N}\}, \qquad \mathcal{A} = [0,1]. \tag{21}$$

Assign state coordinates

$$\eta(s_0) = 0, \tag{22}$$

$$\eta(g_\infty) = \tfrac{1}{3}, \qquad \eta(g_n) = \tfrac{1}{3} + \frac{1}{100(n+1)}, \tag{23}$$

$$\eta(b_\infty) = \tfrac{2}{3}, \qquad \eta(b_n) = \tfrac{2}{3} + \frac{1}{100(n+1)}. \tag{24}$$

Embed $z = (s,a)$ as $\iota(z) = (a, \eta(s)) \in [0,1]^2$. The image $\iota(\mathcal{Z})$ is compact because both limit-state lines are included. Let κ_ν be the unit-variance Matérn kernel of smoothness $\nu > 1/2$ on $[0,1]^2$ and define

$$k(z,z') = \kappa_\nu(\iota(z), \iota(z')). \tag{25}$$

The embedding is injective, so this is simply the restriction of a fixed Matérn kernel to a compact subdomain.

Choose $q \in C^\infty([0,1])$ that equals one on the set of good-state coordinates and zero on the initial and bad-state coordinates. Such a function exists because these compact sets are separated. The function $Q(a,y) = q(y)$ has a smooth compactly supported extension to $\mathbb{R}^2$. Matérn RKHSs on bounded Lipschitz domains are norm equivalent to Sobolev spaces of finite order (Wendland, 2005; Kanagawa et al., 2018). Hence the restriction of Q to $\iota(\mathcal{Z})$ has a finite RKHS norm C_q. Set

$$\alpha = \min\{1, C_q^{-1}\} > 0, \qquad r_1 \equiv 0, \qquad r_2(s,a) = \alpha q(\eta(s)). \tag{26}$$

The reward is fixed, known, in $[0,1]$, and has RKHS norm at most one. This reward smoothness is not needed by the open problem, but shows the lower bound survives the reward part of the stronger coordinate assumptions used elsewhere.

C.2 The transition family

Fix $T \geq 2$, set $M = 2T$, and choose pairwise disjoint open intervals $I_1, \ldots, I_M \subset (0,1)$. Let $\psi_j \in C_c^\infty(I_j)$ satisfy $0 \leq \psi_j \leq 1$ and $\max_x \psi_j(x) = 1$. Let $\Delta = 1/4$.

Choose a smooth cutoff $\chi : [0,1] \to [0,1]$ that equals one at zero and equals zero on every good and bad state coordinate. For $j \in [M]$, define the smooth ambient functions

$$F_{j,+}(a,y) = \frac{1}{2} + \Delta\chi(y)\psi_j(a), \qquad F_{j,-}(a,y) = \frac{1}{2} - \Delta\chi(y)\psi_j(a). \tag{27}$$

Their restrictions to $\iota(\mathcal{Z})$ have finite $\mathcal{H}_k$ norms. Put

$$A_T = \max\left\{ \max_{j\in[M],\, \sigma\in\{+,-\}} \|F_{j,\sigma}|_{\iota(\mathcal{Z})}\|_{\mathcal{H}_k}, \tfrac{1}{2}\|1\|_{\mathcal{H}_k} \right\} < \infty \tag{28}$$

and choose the integer

$$N_T \geq A_T/u. \tag{29}$$

The stage-one transitions of instance j are, for every $z = (s,a)$,

$$P_{1,j}(g_n \mid z) = \frac{F_{j,+}(a, \eta(s))}{N_T}, \qquad n \leq N_T, \tag{30}$$

$$P_{1,j}(b_n \mid z) = \frac{F_{j,-}(a, \eta(s))}{N_T}, \qquad n \leq N_T. \tag{31}$$

All remaining target states have probability zero. The two displayed families sum to one and are nonnegative. At stage two, let the transition be independent of z and uniform on $\{g_n, b_n : n \leq N_T\}$. These final transitions do not affect return, but they make the coordinate assumption valid for every $h \in [H]$ under the convention that P_H is specified.

Equations (27)–(29) give

$$\|P_{h,j}(s' \mid \cdot)\|_{\mathcal{H}_k} \leq u \qquad \text{for every } h \in \{1,2\},\ s' \in \mathcal{S}. \tag{32}$$

When starting at s_0, the total probability of a good terminal state is

$$\sum_{n \leq N_T} P_{1,j}(g_n \mid (s_0, a)) = \frac{1}{2} + \Delta\psi_j(a). \tag{33}$$

The episode's expected return is therefore $\alpha(1/2 + \Delta\psi_j(a))$, whose optimum is $\alpha(1/2 + \Delta)$.

The flat instance 0 replaces every ψ_j by zero and uses the same N_T. Conditioned on "good" or "bad", the observed index n is uniform on $[N_T]$ under every instance. Thus the label contains no information beyond the Bernoulli outcome.

C.3 Coupling and regret

Fix an arbitrary randomized history-dependent algorithm A. Under the flat instance, let A_t be its first-stage action in episode t. For each j, define

$$\tau_j = \inf\{t \leq T : A_t \in \operatorname{supp}(\psi_j)\}, \tag{34}$$

with $\inf \varnothing = \infty$. For every realized trajectory of actions,

$$\sum_{j=1}^{M} \mathbf{1}\{\tau_j \leq T\} \leq T \tag{35}$$

because the supports are disjoint. Taking expectations under the flat instance and dividing by $M = 2T$ gives

$$\frac{1}{M} \sum_{j=1}^{M} \Pr_0(\tau_j \leq T) \leq \frac{1}{2}. \tag{36}$$

For a fixed j, couple instance j and the flat instance with the same algorithm randomness and the same random bits used to draw terminal outcomes. Before an action enters $\operatorname{supp}(\psi_j)$, the good probability is $1/2$ in both instances, and the conditional clone label is uniform in both. Their histories and next actions consequently agree until entry. Entry is determined by the action before the new outcome is observed. Therefore

$$\Pr_j(\tau_j \leq T) = \Pr_0(\tau_j \leq T). \tag{37}$$

Equation (36) supplies some j^* with $\Pr_{j^*}(\tau_{j^*} > T) \geq 1/2$. On this event every selected action has expected gap $\alpha\Delta$. The expected realized pseudo-regret equals the standard value regret by the tower property, so

$$\mathbb{E}_{j^*} R_T = \mathbb{E}_{j^*} \sum_{t=1}^{T} \alpha\Delta(1 - \psi_{j^*}(A_t)) \geq \frac{\alpha\Delta}{2} T. \tag{38}$$

The argument already includes the algorithm's private randomness and proves the minimax claim with $c_\nu = \alpha\Delta/2$.

Finally, every Gram matrix of k is a Gram matrix of the ambient Matérn kernel on $[0,1]^2$. Hence

$$\gamma_T(k; 1) \leq \gamma_T(\kappa_\nu; 1) = \widetilde{O}\Big(T^{2/(2\nu+2)}\Big) = o(T), \tag{39}$$

using the standard Matérn information-gain bound (Vakili et al., 2021). This completes the proof of Theorem 3.

D Proofs and separations for Bellman radii

D.1 Proof of Theorem 4

For any state y, take $V = \mathbf{1}_{\{y\}}$ in equation (8). This gives $\mathfrak{C} \le \mathfrak{B}_+$. The inclusion of the positive unit cube in the signed unit ball gives $\mathfrak{B}_+ \le \mathfrak{B}_\pm$. For any V with $\|V\|_\infty \le 1$, write $V = V_+ - V_-$, where $0 \le V_+, V_- \le 1$. Then

$$\|PV\|_{\mathcal{H}_k} \le \|PV_+\|_{\mathcal{H}_k} + \|PV_-\|_{\mathcal{H}_k} \le 2\mathfrak{B}_+(P;\mathcal{H}_k). \tag{40}$$

For $0 \le V \le 1$ on a countable space, the triangle inequality gives

$$\|PV\|_{\mathcal{H}_k} = \left\| \sum_y V(y) p_y \right\|_{\mathcal{H}_k} \le \sum_y V(y) \|p_y\|_{\mathcal{H}_k} \le \mathfrak{V}(P;\mathcal{H}_k). \tag{41}$$

Taking suprema proves equation (10).

For cloning, let $\bar{V}(y) = m_y^{-1} \sum_{i=1}^{m_y} \widetilde{V}(y,i)$. Direct substitution gives

$$\widetilde{P}\widetilde{V} = P\bar{V}. \tag{42}$$

The average of a positive or signed unit-ball function is in the corresponding unit ball. Conversely, every original function has a constant-on-fibers lift. The suprema defining both radii are equal. Also

$$\mathfrak{V}(\widetilde{P};\widetilde{\mathcal{H}_k}) = \sum_y \sum_{i=1}^{m_y} \frac{\|p_y\|_{\mathcal{H}_k}}{m_y} = \mathfrak{V}(P;\mathcal{H}_k). \tag{43}$$

For an $\mathcal{H}_k$-valued countably additive measure $\mathsf{m}(A) = P(A \mid \cdot)$, its semivariation on $\mathcal{S}$ can be written

$$\|\mathsf{m}\|_{\mathrm{sv}}(\mathcal{S}) = \sup_{\|V\|_\infty \le 1} \left\| \int_{\mathcal{S}} V \, d\mathsf{m} \right\|_{\mathcal{H}_k}. \tag{44}$$

This is exactly $\mathfrak{B}_\pm$. On countable spaces, countable additivity in $\mathcal{H}_k$ follows whenever the slice variation is finite. More generally, the identity applies whenever the transition set function is an $\mathcal{H}_k$-valued vector measure. The definition of $\mathfrak{B}_+$ directly proves its minimal-constant claim.

D.2 No dimension-free reverse inequalities

First let the current domain contain one point and let its RKHS be $\mathbb{R}$ with the usual norm. A transition uniform on n next states has

$$\mathfrak{C} = 1/n, \qquad \mathfrak{B}_+ = \mathfrak{B}_\pm = \mathfrak{V} = 1. \tag{45}$$

Thus $\mathfrak{B}_+/\mathfrak{C} = n$.

Second let the current domain be $[n]$ with the Kronecker kernel $k(i,j) = \mathbf{1}\{i=j\}$. Let the transition be deterministic, sending current point i to next state i. Its coordinate vectors form the Euclidean basis, so

$$\mathfrak{C} = 1, \qquad \mathfrak{B}_+ = \mathfrak{B}_\pm = \sqrt{n}, \qquad \mathfrak{V} = n. \tag{46}$$

Hence both remaining gaps can diverge.

D.3 Proof of Proposition 5

For $0 \le V \le 1$, Bochner integrability and the triangle inequality imply

$$\|PV\|_{\mathcal{H}_k} = \left\| \int_{\mathcal{S}} V(s') p(s' \mid \cdot) \, d\mu(s') \right\|_{\mathcal{H}_k} \tag{47}$$

$$\le \int_{\mathcal{S}} V(s') \|p(s' \mid \cdot)\|_{\mathcal{H}_k} \, d\mu(s') \tag{48}$$

$$\le \mathfrak{V}_\mu(P;\mathcal{H}_k). \tag{49}$$

If the pointwise norm is at most u, the last expression is at most $u\mu(\mathcal{S})$.

Suppose $d\mu' = w\,d\mu$ with $w > 0$ almost everywhere. The transition density relative to μ' is $p' = p/w$, so

$$\int \|p'(s' \mid \cdot)\|_{\mathcal{H}_k}\, d\mu'(s') = \int \frac{\|p(s' \mid \cdot)\|_{\mathcal{H}_k}}{w(s')} w(s')\, d\mu(s') = \mathfrak{V}_\mu(P; \mathcal{H}_k). \tag{50}$$

A measurable bijective relabeling gives the same identity by change of variables.

D.4 Proof of Corollary 6

For $h = H$, the claim is immediate because $V = 0$. If $h < H$ and $0 \le V \le H - h$, then $V/(H-h)$ belongs to the positive unit ball. Therefore

$$\|P_h V\|_{\mathcal{H}_k} \le (H-h)\mathfrak{B}_+(P_h; \mathcal{H}_k). \tag{51}$$

The triangle inequality after adding r_h proves equation (12). Setting $r_h = 0$ and taking a supremum over V proves minimality.

E Detailed proof-level literature audit

This appendix records the exact distinctions summarized in Table 1. It is not a claim that every theorem in a cited paper fails. The issue is one implication used to instantiate a Bellman-image premise.

Direct Bellman closure. Yang et al. (2020, Assumption 4.1) assumes that the Bellman operator maps every bounded action-value function into a fixed RKHS ball. Their main kernel theorem is stated under that assumption and is not affected. Immediately afterward, their equation (4.1) gives coordinate transition slices in the unit ball as a sufficient condition and asserts an $H + 1$ target norm. Theorem 4 shows that this sufficiency statement needs a bound on $\mathfrak{B}_+$, slice variation, or state volume.

The source of the volume term. Yeh et al. (2023, Assumption 2) assumes a pointwise RKHS norm bound for the transition density. Their Lemma 3 proves

$$\|PV\|_{\mathcal{H}_k} \le \frac{c}{1-\gamma}, \qquad c = \int_{\mathcal{S}} ds', \tag{52}$$

for discounted values bounded by $1/(1-\gamma)$. Their appendix proof displays the integral and the factor c. This agrees with Proposition 5. The lemma is valid when the density convention and finite c are part of the model.

This distinction also identifies a valid normalized special case. Vakili et al. (2024, Lemma 2.3) states its value-function domain as $[0,1]^d$ and invokes the same density calculation. Under the stated Lebesgue unit-cube convention, $c = 1$, so the displayed radius-H conclusion retains the entire volume factor. An extension to other regular compact domains, as suggested later in that paper, must retain their volume or an invariant integrated bound.

Dimension-free uses. Vakili & Olkhovskaya (2023, Assumption 1 and Lemma 1) assumes unit coordinate norms and calls membership of every $r_h + P_h V$ in the radius-$(H+1)$ ball an immediate consequence, citing the preceding lemma. The cited proof contains c. Thus the dimension-free consequence is not established on the paper's general state space. Theorem 3 applies directly to the transition-only Assumption 1 posed by Vakili (2024).

Vakili & Olkhovskaya (2024, Assumptions 2 and 4) separately assumes coordinate transition norms and optimistic closure for value functions in a state RKHS. In the proof of its Theorem 3, the required state-action RKHS bound $\|Pv\|_{\mathcal{H}_k} = O(w)$ is then obtained from the coordinate premise by citing Yeh et al.'s Lemma 3. Because its state space is general, that step requires the omitted reference-volume or integrated-variation constant. The paper's self-contained confidence theorem is stated directly with a bound on $\|Pv\|_{\mathcal{H}_k}$ and is not affected by this point.

Kayal et al. (2025, Assumption 1) likewise states $\|PV\|_{\mathcal{H}_k} = O(H)$ and cites Yeh et al. The statement is correct only if the suppressed constant includes the reference volume or an integrated variation bound. The same distinction applies to the coordinate premise and imported KOVI bound in Cioba et al. (2026), and to the arbitrary-policy Q-norm claim following equation (4) of Pavlovic et al. (2026). These are recent preprints or applications with additional assumptions. We make no claim here about parts of their arguments that do not use the coordinate-to-Bellman step.

Independent issues and invariant alternatives. Appendix F of Maran et al. (2024a) observes that a cover restricted to a current partition cell does not control values at unrestricted next states in Vakili & Olkhovskaya (2023). That moving-target localization issue is independent of the coordinate-volume obstruction proved here. The smooth-Bellman assumptions and local linearity results of Maran et al. (2024a;b) directly control Bellman images and survive cloning.

Chowdhury & Oliveira (2023) assumes an operator-level conditional mean embedding that maps a chosen state RKHS into the state-action RKHS, together with closure of the algorithm's optimistic value class. Such operator assumptions are invariant under consistent pullbacks. They are not identical to $\mathfrak{B}_+$, since their domain is a specified state RKHS rather than all bounded functions. The measure-theoretic CME literature carefully distinguishes these representability conditions (Park & Muandet, 2020; Hertel et al., 2026).

The 2025 OpenReview preprint of Kumar (2025) also assumes a bounded conditional mean embedding and projects optimistic proxies into a chosen state-RKHS ball. This is a recent algorithmic use of the invariant operator route. It neither derives Bellman closure from coordinate transition slices nor studies state cloning, so its contribution is complementary to ours.

F Verification artifact

The file `audit_cloning.py` performs four independent finite checks.

1. It samples a transition matrix on a finite current domain, computes exact finite-RKHS coordinate norms, and verifies their prescribed clone scaling.
2. It enumerates the vertices of the positive and signed value cubes to compute both Bellman radii. It verifies that they are unchanged after nonuniform cloning.
3. It verifies invariance of slice variation and equality of original and pulled-back information gains for many point sequences.
4. It exhaustively checks the support-count inequality (35) for every length-four sequence over eight supports and verifies the resulting one-half no-hit fraction.

All comparisons use a tolerance of 10^{-10}. These checks are not substitutes for the proofs. They guard against indexing, scaling, and inequality-direction errors.